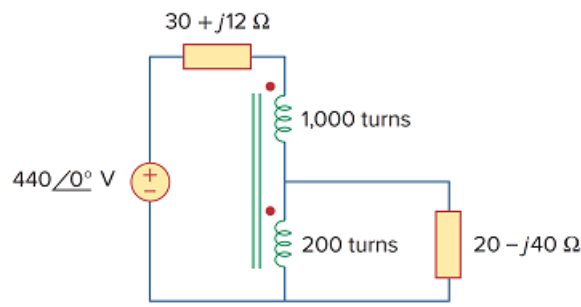


Problem

In the ideal transformer circuit shown in Fig. 13.133, determine the average power delivered to the load.

Figure 13.133



Step-by-step solution

Step 1 of 7

Refer to Figure 13.133 from the textbook.

Observe that the circuit given is a step-down auto transformer.

Assume the direction of the currents, I_1 and I_2 as shown in Figure 1.

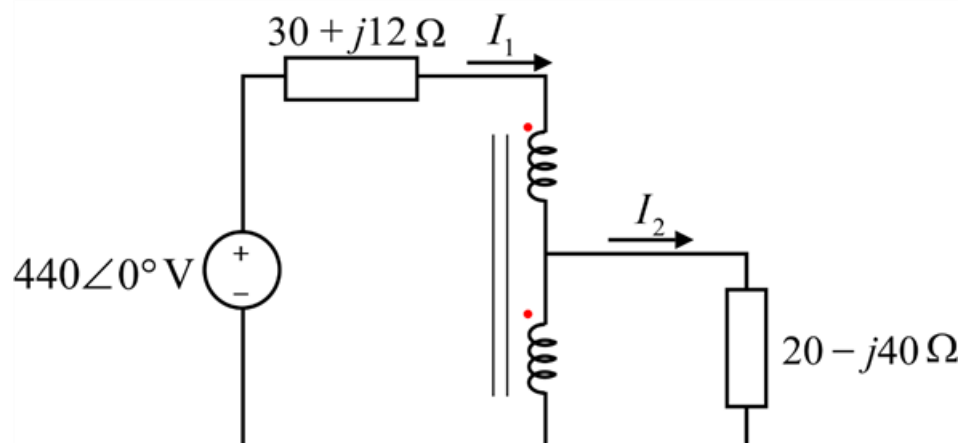


Figure 1

Step 2 of 7

For a step-down autotransformer,

$$\frac{V_1}{V_2} = \frac{N_1 + N_2}{N_2} \dots\dots (1)$$

and

$$\frac{I_2}{I_1} = \frac{N_1 + N_2}{N_2} \dots\dots (2)$$

Here,

V_1 is the primary side voltage,

V_2 is the secondary side voltage,

N_1 is the number of series winding turns,

N_2 is the number of common winding turns.

Step 3 of 7

Substitute 1000 for N_1 and 200 for N_2 in equation (1).

$$\frac{V_1}{V_2} = \frac{200 + 1000}{200}$$

$$\frac{V_1}{V_2} = \frac{1200}{200}$$

$$\frac{V_1}{V_2} = 6$$

$$V_1 = 6V_2$$

Step 4 of 7

Substitute 1000 for N_1 and 200 for N_2 in equation (2).

$$\frac{I_2}{I_1} = \frac{1000 + 200}{200}$$

$$\frac{I_2}{I_1} = \frac{1200}{200}$$

$$\frac{I_2}{I_1} = 6$$

$$I_1 = \frac{1}{6}I_2$$

Step 5 of 7

Apply the Kirchhoff's Voltage Law in the loop ABEF.

$$440\angle 0^\circ = (30 + j12)I_1 + V_1 \dots\dots (3)$$

Apply the Kirchhoff's Voltage Law in the loop BCDE.

$$V_2 = (20 - j40)I_2 \dots\dots (4)$$

Step 6 of 7

Substitute $\frac{1}{6}I_2$ for I_1 and $6V_2$ for V_1 in equation (3).

$$440\angle 0^\circ = (30 + j12)\left(\frac{1}{6}I_2\right) + 6V_2$$

$$440\angle 0^\circ = (5 + j2)I_2 + 6V_2$$

Substitute $(20 - j40)I_2$ for I_2 .

$$440\angle 0^\circ = (5 + j2)I_2 + 6((20 - j40)I_2)$$

$$440\angle 0^\circ = (5 + j2)I_2 + (120 - j240)I_2$$

$$440\angle 0^\circ = (125 - j238)I_2$$

$$I_2 = \frac{440\angle 0^\circ}{(125 - j238)}$$

$$I_2 = 1.637\angle 62.29^\circ \text{ A}$$

Step 7 of 7

Write the expression for the average power delivered to the load, p .

$$p = |I_2|^2 R$$

Substitute 1.637 A for $|I_2|$ and $20\ \Omega$ for R .

$$p = (1.637)^2 (20)$$

$$= (2.68)(20)$$

$$= 53.6 \text{ W}$$

Therefore, the average power delivered to the load, p is 53.6 W.